OHLALA
CONTENT-COMPLETENESS, COMMUNICATION AND EMOTIONAL EXPERIENCE

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ABSTRACT

Among other explorations, the field of telepresence technology has looked at ways to create a feeling of telepresence based on the transfer of minimal information. On this topic, the Cololo project has taken an extreme position by proposing the experience of 1-bit communication.

Based on the observation of Cololo in use, it is shown that content is not necessary to trigger an emotional experience. This paper introduces a novel dimension to be taken into consideration in communication technology: the content-completeness dimension, ranging from non-content to hyper-content. Furthermore, we built the Ohlala framework, aiming to explore the content-completeness dimension. Based on Ohlala, by way of a research through design, we intend to explore further the relations between this dimension on communication and emotional experience.

Keywords: content-completeness, communication, emotional experience

INTRODUCTION

More, faster, further. Recent developments in digital technology have supported the design of tools that enable us to exchange more information, faster, and further. Nowadays it is feasible to connect entire spaces and people, i.e. to create a feeling of telepresence. However, an alternative solution has also been explored: creating the feeling of telepresence based on the transfer of *minimal information*.

Taking it to an extreme and reducing the information to one bit would prevent transfer of any content (described as a set of information used with the intention of conveying meaning). In this case, what would be the consequences for the feeling of telepresence, and more comprehensively on emotional experience in communication?

This position paper explores the notion of telepresence towards 1-bit communication, and its consequences for emotional experience. Based on the observation of 1-bit communication (by the means of the design project Cololo), this paper introduces a novel dimension to be taken into consideration in communication technology: the content-completeness dimension. Furthermore, this paper presents the framework Ohlala, which is being designed to explore deeper the aforementioned dimension in order to research implications on communication and on emotional experience.

Less (information), slower (technology), closer (emotionally)

TELEPRESENCE

Within the design field of telecommunication, a growing domain is dedicated to telepresence. Telepresence technology aims to create systems to share presence and sometimes a common space between people. Telepresence projects always involve nonverbal communication and paralanguage information, i.e., not what one says, but how one says it (A. Kaye, 1977). Our aim in this section is to explore how telepresence technology has lead toward communication opportunities based on minimal information transfer.

Two main directions are followed by telepresence projects: explicit and implicit telepresence. 'Explicit telepresence' involves great amounts of information (such as video, sound, possibly haptic technology) to enable people to have a similar communication experience - same quality and almost the same capabilities – to what they would have if they were actually in the same place (e.g. (Cisco, 2011; Telecom France, 2011)). 'Implicit telepresence' aims to look for minimal solutions (towards a minimum amount of information; one, or at most two modalities; etc.) for the sharing of presence, of emotions, of subtle information, or of simple remote actions (e.g for tangible designs (Itoh, Miyajima, & Watanabe, 2002; Suzuki & Hashimoto, 2004), and for computer based ones (J. "Jofish" Kaye, Levitt, Nevins, Golden, & Schmidt, 2005)).
In this paper, our focus is implicit telepresence.

In this group of telepresence projects, I/O devices are our focus. Suzuki & Hashimoto (2004) propose the following key factors of bidirectional I/O device:

- Simplicity: A simple communication method with a simple system that should be enough small to carry on. […]
- Affordability: From the viewpoint of affordance and embodiment of the interface, it must be easy for users to imagine how they can use the device. Affordance is a relationship between the object and the task by the object.
- Synchronicity: Regardless of immediate or delayed communication, a device must enable people to feel a sense of connectedness in an intuitive manner.
- Creativity: The usage of device should not be limited so that users can find out a new use of the device expansively. Users build their own rule and communication language like a play game. This emerges a new communication protocol between users' local community.

In order to clarify our focus in the field of telepresence and to precise the context of our project, we propose here a reflection on a few projects which, among many, have caught our attention in particular.

SyncDecor (Tsujita, Tsukada, & Siio, 2007) aims at bringing telepresence devices in the home of distant couples. Actions in one place may have similar consequences in the remote place (i.e. switching a light may also switch a light in the remote place). Therefore, SyncDecor aims at remotely applying feedforwards (S. A. G. Wensveen, Djajadiningrat, & C. J. Overbeeke, 2004) to create the impression of telepresence, which seems quite intrusive from a design point of view. For example, if the SyncLamp (a set of lamps based on the SyncDecor technology, synchronizing the brightness of two connected lamps) is turned on at maximum in one place, then the remotely connected lamp would also light up at its maximum, which may interfere with the ambiance of calmness wished by the distant person. From a design point of view, the risk shown here is that such intrusive technologies for telepresence may be contextually inappropriate, and therefore wrongly experienced.

The Bed (Dodge, 1997) proposes to contextualize the use of telepresence technology for a couple at a "familiar and intimate space": the bed. This contextualization of telepresence to a specific space is indeed a wise idea to reduce the possible negative impression by intrusion, but also to limit and to constraint the time of use.

Another solution, aiming at less intrusion, is proposed by Tsunagari (Itoh et al., 2002). The presence and the motion of one person are remotely expressed respectively by light and rotation of the Tsunagari artifact (a planter). This is less intrusive as most likely it would not disturb the activity of the distant person. However, just like LumiTouch (Chang, Resner, Koerner, Wang, & Ishii, 2001), a picture frame conveying haptic pressure into light to a distantly connected picture frame, Tsunagari is based on a non-symmetrical modal structure, with two different modalities between expression (touch) and perception (sight). As the actual means of expression is not rendered identically, this can be seen as a partial loss of information. However, the telepresence can still be perceived positively, and causes fewer problems due to non-intrusive expression.

The InTouch project (Brave & Dahley, 1997) proposes a symmetrical modality (touch) for interpersonal communication, involving haptic feedback technology. The use of touch and physical contact are seen as "a basic means through which people achieve a sense of connection, indicate intention, and express emotion." Through bi-directional "haptically augmented" devices, connected users can feel the presence of each other, and act together to control speed and direction of any of the three rollers.

The device presented by Lenay (2010) also takes into consideration the strength of touch in the sharing of emotions. To interact with a distant person (using the device as well), the subject acts on a computer mouse to move a cursor on a 600-pixel segment. When both cursors meet each other (or with a lure) on the segment, an all-or-nothing tactile stimulus is sent to the index of the free hand of the subject. Based on a minimal setup (avoiding complexity as much as possible) to experiment the phenomenon of distant touch, Lenay has figured out fundamental knowledge to understand perceptual crossing (Deckers, S. Wensveen, Ahn, & K. Overbeeke, 2011). This knowledge forms a valuable ground to design from, but is not the direct purpose of Lenay.

Minimal qualities were also a concern of the Cololo project (Ogaki, Suzuki, Hoshikawa, & Uchiyama, 2008), but with a design intention. We focus on this project in the next section.

COLOLO

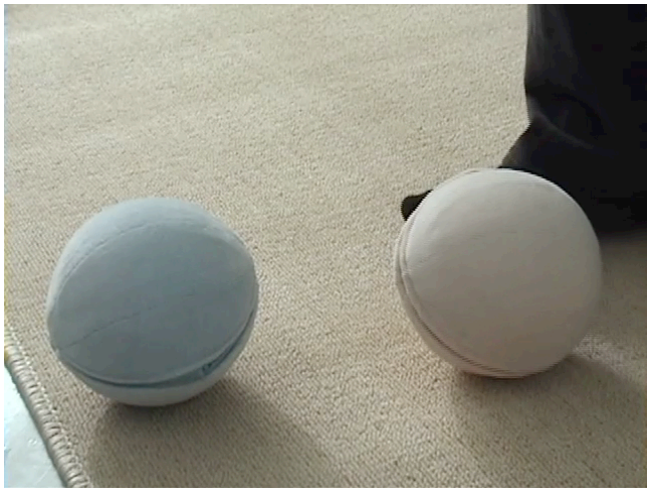


Figure 1. Picture of the Cololo project.

INTRODUCTION TO COLOLO

The Cololo project (Ogaki et al., 2008) proposes a way of experiencing 1-bit communication.

As shown in Figure 1, Cololo is simply a set of two balls connected to each other wirelessly (through the internet). If one ball (we will call it "the first ball") is being moved, then the other one (we will call it "the second ball") moves too, and vice versa. Yet the movement of the second ball is random. That is, the qualities of the movement are not transferred. Therefore, seeing the second ball moving provides only the clue that the first ball has moved. No information is conveyed concerning the reason of the movement, nor concerning the quality of the movement. This means of communication is therefore truly a 1-bit communication, the 1-bit of information being "the first ball has moved".

Cololo can thus be described as a telepresence device based on 1-bit communication. From the second ball, any user may apprehend the presence of the person who has moved the first ball. Note that even if the first ball has moved for any other reason (an animal moving it, wind…), the user owning the second ball might believe in the intention of the distant person to move the first ball, and therefore experience a feeling of telepresence of the other (this fact may happen with any I/O telepresence device).

In any situation of use, no content can be transferred between the two users connected through Cololo. No message can be expressed. No language can be created nor used. Yet, each time the second ball moves, we notice that the observer reacts emotionally. Therefore the intention of the expresser, who moves the first ball, triggers an emotional reaction in the observer, by the movement of the second ball. Although the 1-bit communication technology does not transfer any content, it triggers an emotional reaction. This observation has previously been reported as well by the research work on the 1-bit communication device called Virtual Intimate Object, VIO: "We feel these instances of competition emphasize the situated nature of communication through the VIO and underline the rich potential for interpretation even

in a 1-bitcommunication device." (J. "Jofish" Kaye et al., 2005)

REFLECTION

Reflecting on the Cololo project and observing the use of Cololo, we can state a few points.
From the point-of-view of the observer, the perception of the ball in movement does create an emotional experience. Involving expectation and contextual knowledge, such as the relationship with the person using the first ball, the movement of the second ball triggers an emotional reaction in the observer. Therefore, we pose here that content is not necessary to trigger an emotional experience.

This opens up a new direction in the domain of communication studies. In recent years, much research has been done to describe and to find applications of the implicit/explicit content dimension (e.g. (Rauterberg, 2006) on cultural computing). This dimension is of course fundamental to the field of (non-verbal) communication. Yet, Cololo points to another dimension, which is our main research topic here: the content-completeness dimension, ranging from non-content to hyper-content (cf. Figure 2). "Hyper-content", as we use it, means that similar content is expressed and perceived redundantly.

On one extreme of this dimension, Cololo is a communication device with no content. On the other extreme, synesthetic experience simulators (e.g. the colourful-rain umbrella (Lévy, Kim, Tsai, Lee, & Yamanaka, 2009)) can be seen as hyper-content projects. The colourful-rain umbrella lets its user to experience a synaesthetic perceptive phenomenon: all sounds in the rainy street are also perceived as colors on the canopy.

In between, many projects and objects might be placed according to the completeness of the information. For example, the research done by Emberson et al. (2010) on halfalogues can be placed on this scale. Halfalogues are what one hears from somebody else's cell phone conversation: half of what is being said. They are shown to be distracting, and even potentially dangerous.

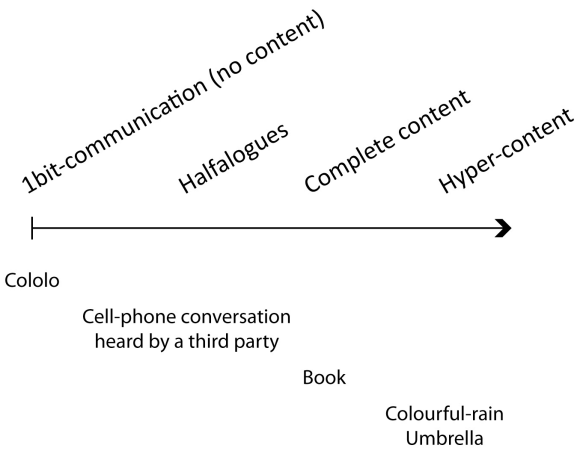


Figure 2. The dimension of content-completeness, with examples.

Finally, it is interesting to notice that the content-completeness dimension is independent from the implicit/explicit one. Indeed, this dimension does not take into consideration the qualities of the content (implicit/explicit) nor the modality to express or capture it (sound, light, haptic…). We believe that this dimension is also suggested by Marshall McLuhan's seminal quote "the medium is the message" (M. Mcluhan, 1964).

DISCUSSION

Through the design of Cololo, an important point has been raised: emotional experience is triggered not only by content. Although Cololo does not convey any content between the two balls, the user of the second ball does have an emotional experience triggered by the movement of this ball. However, what is described here does relate to emotional arousal only, but not to emotional valence (Dolcos, LaBar, & Cabeza, 2004).

The authors could not find any relevant literature on the topic (i.e. the relation between content-completeness and arousal of emotional experience). There is mention of minimal requirements of media to establish believable 'humanness'. However this is in the context of HCI and not in the context of telepresence (Reeves and Nass, 1996). Furthermore, Cololo cannot help to understand or even give any clue as to any relation between the content-completeness and the level of arousal in the emotional experience. What is shown is the elicitation of an emotional experience, but there is no understanding about the intensity of the emotion.

Moreover, it should be noticed that "complete content" media, such as books, can influence the valence of the emotional experience. In the case of 1-bit-communication media, Cololo is not capable to direct such influence on the observer. In regard to this point, there is some literature on "partial-complete content" (Emberson et al., 2010). However, much remains to be explored on the relation between emotional experience and content-completeness.

To summarize, the design of Cololo has opened up several research questions to be explored through and for design:

- What is the impact of information completeness on emotional experience?
- How to design for emotional experience?
- As content is not necessary for emotional experience, what is?

Especially the last point is a challenging topic in the field of Kansei studies (Lévy, Nakamori, & Yamanaka, 2008), as it implies a research exploration of non-informational based perception and experience.

To explore through research these points shown by the Cololo project, the authors propose to create a research framework called Ohlala, aiming at creating different case studies on the dimension of content-completeness.

OHLALA FRAMEWORK

INTRODUCTION

From the Cololo project, positioned at one extreme of the content-completeness dimension, it was found that emotional experience might be triggered by something else than content. We will call this the "triggering force". This triggering force resides within the perceptual activity that takes place between the second ball and its observer.

To explore further the effects of this triggering force, and of the content completeness in the emotional experience, we propose to create a framework allowing the experience of various levels of content-completeness situations, from 1-bit communication to complete and rich content-based communication experience. We also plan to integrate hyper-content in a later iteration of the Ohlala framework design.

Based on the categories of tangible user interfaces described by Ishii (2008), Ohlala appears to be a framework exploring a broad range in the field of tangible telepresence. Thanks to the implementation of our framework, we are able to explore various factors influencing the content-completeness dimension through communication, and their impact on perceptive experience, in both emotional and cognitive domains.

STRUCTURE

For the implementation of the Ohlala framework, we start from Cololo, i.e. from a minimalistic point of view. Then we extend the implementation to enable the exploration of the entire range of the content-completeness dimension.

The basic structure of Ohlala is described in the following. In a later section we discuss several implementations of this structure that are currently being developed.

Each participant using Ohlala is connected to a cursor, being a contact element of a node of Ohlala. The cursor can be moved in any direction. For now, movements range from one dimensional to two-dimensional, plus a rotation of the cursor on the axis orthogonal on the surface. When one of the cursors is moved by a participant, any other connected cursor moves as a reaction. However, the way the reacting cursors move depends on the *transformative behaviors* (cf. next section) between the two connected cursors. For example, in the case of Cololo, the *transformative behavior* can be described as a "delayed random movement". So, to move the second ball based on the first ball, Cololo provoked a delayed random movement.

Whereas Cololo is designed for a one-to-one communication, with Ohlala we aim at keeping open the possibility of extending the framework to afford for groups of individuals communicating together. In

such networks, one cursor (node in the Ohlala network) influences one or more cursors, and inversely, a cursor is influenced by one or more other cursors in the network.

TRANSFORMATIVE BEHAVIORS

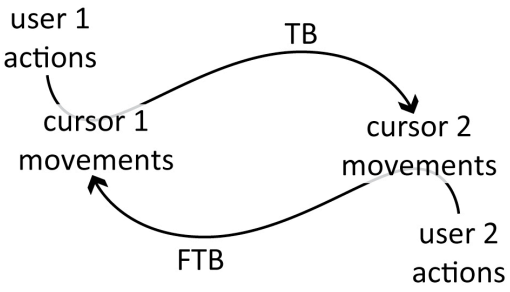


Figure 3. Transformative behaviors

A transformative behavior describes the way the movements of one cursor of Ohlala influence connected cursors' movements. Figure 3 shows the way transformative behaviors (TB) are impacting the movement of a cursor (cursor 2 in this case).

In the simplest case (cursor 1 – controlled by user 1 – affects cursor 2 – controlled by user 2), the movements of cursor 1 provoked by user 1 actions are translated into a TB. This TB impacts the actual movements of cursor 2.

For clarity's sake we here describe the system as seen from user 1 only. Obviously the description mirrors when seen from user 2.

Considering the effect of cursor 2 on cursor 1, two TB cases are possible:

- A two-way communication with inhibited feedback – This means that while cursor 1 affects cursor 2, cursor 2 cannot affect cursor 1. They can affect each other, but not at the same time.
- A two-way communication with continuous feedback – Each cursor can affect the other one at any moment, even at the same time. The feedback is about how much user 2 actions affect the cursor 2 movements provoked by the TB, i.e. how they mesh together. Although the feedback technically is a TB as well, we call it feedback transformative behavior (FTB) when described from the point of view of the user 1.

All Ohlala TB can be defined by the combination of four qualities (cf. Table 1):

- Time delay: how long after the movement of the first cursor the second one will move. It can be instant (no delay – case D1), with a fixed delay (case D2), with a random delay (case D3).
- Speed (temporal scaling): how much faster or slower the second cursor moves compared to the first cursor. It can be the same (case L1); the ratio between speeds of each cursor can be fixed (case L2) or random (case L3).
- Space modification (spatial scaling or distortion): how the second cursor moves in space, relative to the movements of the first cursor. The movement could be the same (duplication of the movement – case S1), proportional (i.e. same pattern but on a smaller/greater scale – case S2), random but with the same distance covered (case S3), random but finishing at the same point for each continuous movement (case S4), mix of S3 and S4 (case S5), similar but with a different angle (case S6), completely random (case S7).
- Feedback conditions: As described previously, a two-way communication can be with inhibited feedback (case F1) or with continuous feedback (case F2). A random switch between these two cases is also possible (case F3).

Quality	Type	Case number
Time delay	No delay (instant)	D1
	Fixed delay	D2
	Random delay	D3
Speed	Similar (no modification)	L1
	Fixed ratio (cursor 1/cursor 2)	L2
	Random ratio	L3
Space modification	Similar movement	S1
	Proportional movement	S2
	Same distance covered	S3
	Same finish point	S4
	S3 & S4	S5
	Angle modification	S6
	Complete random movement	S7
Feedback condition	Inhibited feedback	F1
	Continuous feedback	F2
	Random condition	F3

Table 1. TB qualities between the first and the second cursor

Note that Table 1 does not list all possible ranges of the domains completely. However, starting from the minimalistic point of view, these are the ones

currently implemented in Ohlala. Applying these qualities defines Cololo as {D3,L3,S7,F1}.

DESIGN ITERATIONS

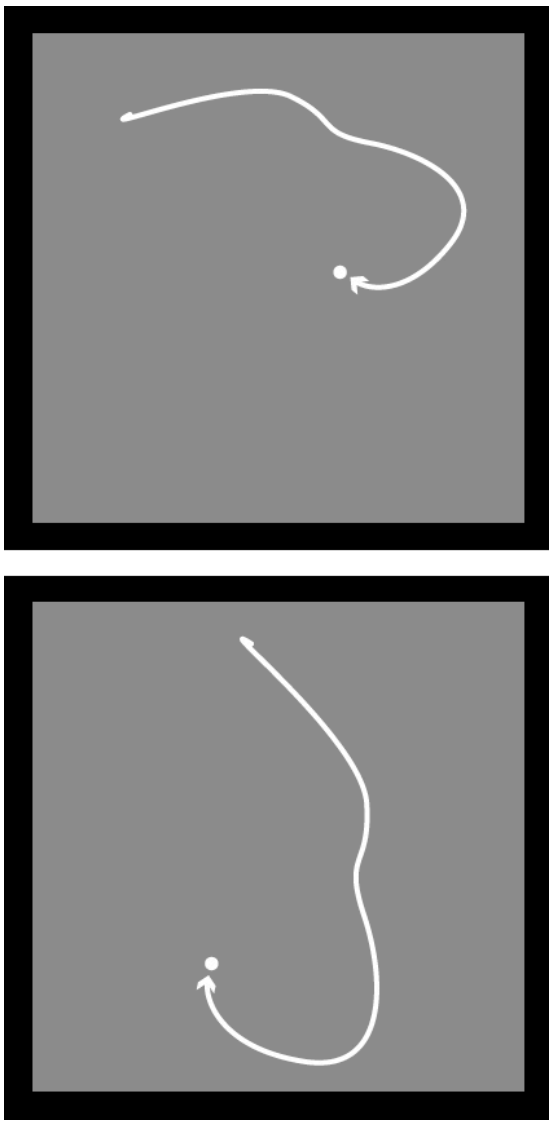


Figure 4. Screenshots of the first iteration of the design of Ohlala. The upper square shows the screen of the user 1. The lower square shows the screen of the user 2. For each screen, the white point is the cursor, which can move only in the grey zone. The movement of the cursor is represented by the white curved arrow. In this example, the movement of the cursor 1 is provoked by the user 1, and the one of the cursor 2 by the BT between cursor 1 and cursor 2, The BT can be described as {D1,L2, S3-S6,F1}. Therefore the movement the cursor 2 is relatively proportionally bigger and with an angle.

The first iteration in the design of the Ohlala framework consists in implementing the various TB previously described. Behaviors are first implemented and simulated on screen. Figure 4 shows two screens of this iteration. The top square is the screen of the cursor 1, controlled and seen by the user 1. The bottom square is the screen of the cursor 2, controlled and seen by the user 2. Users 1 and 2 control their cursors from two different computers. Each cursor is controlled by a computer mouse. In Figure 4, the two white points show the position of the cursor. The different positions are due to the TB.

Following iterations will aim at the physicalisation of the Ohlala framework. The cursors are planned to be physical artifacts able to move with similar qualities as the cursor does digitally. Each artifact will be controlled by a computer, which can connect to other computers.

Latency, feedforward and feedback are expected to be challenges for the design of all iterations of the Ohlala framework. Sponginess, bumpiness, noise, etc. are expected to be additional challenges for the physical iterations.

CONCLUSION

This paper has presented the reflections and the findings on which the framework Ohlala is based and designed.

Technological developments around minimal information based telepresence have led to our new field of exploration, valuable for interaction design, which is the content-completeness dimension. A little information transferred wisely can trigger a rich emotional experience, when it is perceived as shared between distant yet connected people.

Moreover, reflecting on the Cololo project, we have found the following: in the context of communication, emotional experience can triggered by something else than content. This has led to the description of a new dimension valuable in interaction design: the content-completeness dimension.

These findings are crucial as they lead to fundamental questions for interaction design and kansei design: what is the triggering force eliciting emotional experience? How to design for it? How does content (implicit or explicit) influence these experiences?

We believe that the Ohlala framework supports the exploration of these challenges presented to interaction design and kansei design. Ohlala is expected to help to answer these questions partially, and possibly to find other challenges to be addressed by researching and designing.

Ohlala is expected to enable a deeper understanding of the content-completeness dimension, integrating feedback in the communication. Furthermore, we envision a system configuration (with multiple cursors and complex relations).

Based on the findings of this research, we hope that knowledge and know-how will be gained to design artifacts capable of providing new tools for designing emotional experiences in interaction. As content is shown not to be the essential in these experiences, we also hope to find other means to create and to enhance such experiences.

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